MTH344 Sample Midterm (100 pts + EC 20 pts)

Please print your name clearly. A penalty may be applied if your name is difficult to read.

Last Name: ______________________________

First Name: ______________________________

Instructions:

- There are nine problems. The points for each problem are indicated next to the question.
- To earn full credit, you must show all necessary work. If it is unclear which assumptions or theorems may be used without proof, write out as much of your reasoning as possible.
- This test is closed-notes, and no computing devices (including calculators) are allowed.

P1 (10 pts)
a. (5 pts) Show that, if X is a random variable with finite mean μ and variance σ^2, then for any value $k > 0$,

$$P(|X - \mu| \geq k) \leq \sigma^2/k^2. \tag{1}$$

b. (5 pts) Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$. Let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Then, show that, for any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| > \epsilon) \to 0 \quad \text{as } n \to \infty. \tag{2}$$

P2 (15 pts)

Let $T \sim t(n)$. Show that $E[|T|^k] = \infty$ when $k \geq n$.

P3 (15 pts)

An estimator $\hat{\theta}_n$ for a parameter θ is called *asymptotically unbiased* if $\lim_{n\to\infty} \mathrm{E}[\hat{\theta}_n] = \theta$. Show that if an estimator $\hat{\theta}_n$ is asymptotically unbiased and $\lim_{n\to\infty} \mathrm{Var}[\hat{\theta}_n] = 0$, then the estimator is consistent.

P4 (10 pts)

Let X follow an exponential family distribution whose PDF is given by:

$$f_X(x;\eta) = h(x)\exp\left(\eta T(x) - A(\eta)\right), \tag{3}$$

satisfying the following regularity conditions:

(C1) The natural parameter space $\eta(\Omega)$ is an open interval.

(C2) $T(X)$ is not degenerate. That is, $\mathrm{Var}_\eta[T(X)] > 0$ for all $\eta \in \eta(\Omega)$.

a. (5 pts) Show that $A'(\eta) = \mathrm{E}_\eta[T(X)]$ and $A''(\eta) = \mathrm{Var}_\eta[T(X)]$.

b. (5 pts) Show that $\hat{\eta} \in \eta(\Omega)$ is the unique MLE if it satisfies the following *likelihood equation*:

$$\mathrm{E}_{\hat{\eta}}[T(X)] = \overline{T(X)} := n^{-1}\sum_{i=1}^{n} T(X_i). \tag{4}$$

P5 (10 pts)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Pareto}(\alpha, \beta)$. Its PDF is given by:

$$f(x; \alpha, \beta) = \frac{\alpha \beta^\alpha}{x^{\alpha+1}}, \quad x \geq \beta,$$

and zero otherwise. Here, $\theta = (\alpha, \beta)^T$ and $\Omega = \{(\alpha, \beta) : \alpha > 0, \beta > 0\}$. Find the MLE for θ.

P6 (10 pts)

Using moment-generating functions, show that as $n \to \infty$, $p \to 0$, and $np \to \lambda$, the binomial distribution with parameters n and p tends to the Poisson distribution.

P7 (15 pts)
a. (5 pts) Show that the PDF of $\chi^2(n)$ is given by:

$$f(v) = \frac{1}{2^{n/2}\Gamma(n/2)} v^{(n/2)-1} e^{-v/2}, \quad v \geq 0. \tag{5}$$

b. (10 pts) Show that the PDF of $t(n)$ is given by:

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\,\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2} \tag{6}$$

P8 (15 pts)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where $\theta = (\mu, \sigma^2)^\top$ and $\Omega = (-\infty, \infty) \times (0, \infty)$.

a. (10 pts) Show that its PDF belongs to the multi-parameter exponential family satisfying conditions (C1) and (C2):

(C1) The natural parameter space $\boldsymbol{\eta}(\Omega)$ is an open set in $\mathbb{R}^k$.

(C2) The covariance matrix $\mathrm{Var}_{\boldsymbol{\eta}}[\boldsymbol{T}(\boldsymbol{X})]$ is positive definite (non-singular) for all $\boldsymbol{\eta} \in \boldsymbol{\eta}(\Omega)$.

b. (5 pts) Use the likelihood equation to find the MLE for θ.

P9 (EC 20 pts)
a. (10 pts) Let $(X_n)_{n\geq 1}$ be a sequence of random variables and θ be a constant. Suppose that $\sqrt{n}\,(X_n - \theta) \xrightarrow{d} Z$ for some random variable Z. Then, for any twice differentiable function g with a continuous second derivative such that $g'(\theta) = 0$, show that:

$$n\left(g(X_n) - g(\theta)\right) \xrightarrow{d} \frac{g''(\theta)}{2} Z^2. \tag{7}$$

b. (10 pts) Let X_i be i.i.d. standardized random variables (i.e., $\mu = 0$ and $\sigma^2 = 1$). Show that, for any $0 < \alpha < 0.5$:

$$\lim_{n\to\infty} P\left(1 - \frac{1}{2n}\chi^2_{\alpha/2}(1) \leq \cos\bar{X} \leq 1 - \frac{1}{2n}\chi^2_{1-\alpha/2}(1)\right) = 1 - \alpha. \tag{8}$$